

Volterra Functional Approach to the Poisson Channel with Noisy Feedback: a Note

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Abstract

In this note the authors propose to apply Volterra's approach to the Poisson channel aided by noisy Poisson feedback.

From Wyner [1] we have the definition of the Poisson channel,

The channel input is a waveform $\lambda(t)$, $0 \leq t < \infty$, which satisfies $0 \leq \lambda(t) \leq A$, where the parameter A is the peak power. The waveform $\lambda(\cdot)$ defines a Poisson counting process $v(t)$ with “intensity” or “rate” equal to $\lambda(t) + \lambda_0$, where $\lambda_0 \geq 0$ is a background noise level (sometimes called “dark current”). Thus the process $v(t)$, $0 \leq t < \infty$, is the independent-increments process such that $v(0) = 0$, and, for $0 \leq \tau, t < \infty$,

$$Pr v(t + \tau) - v(t) = j = \frac{e^{-\Lambda} \Lambda^j}{j!}, j = 0, 1, 2, \dots \quad (1)$$

where

$$\Lambda = \int_t^{t+\tau} (\lambda(t') + \lambda_0) dt' \quad (2)$$

Physically, we think of the jumps in $v(\cdot)$ as corresponding to photon arrivals at the receiver. We assume that the receiver has knowledge of $v(t)$, which it would obtain using a photon-detector.

Wyner writes out the decoder considered, in Section IV.A - it is the maximum likelihood decoder. We consider a four phase communication scheme [2, 3] of which the first phase is, as usual, the transmission of the message over the Poisson channel. We focus on the first phase and note where differences may come in when considering the Wiener/Volterra kernel approach [4] as opposed to Wyner's more mainstream approach.

In [5] the authors provide the first three Wiener/Volterra kernels of a Poisson input-output relation. Thus the output of their Poisson black box can be

expressed in terms of these. Suppose one of M messages is transmitted over the black box. We compute the outputs for each of the M possibilities. Then we compute the a posteriori probability (APP) of the output given each of the M possibilities and pick the one with the maximum APP. This computation is carried out on the basis of the first three Wiener/Volterra kernels. By increasing the number of Volterra kernels considered one can perhaps get better accuracy.

Thus, the problem that we are suggesting in this note is to rederive Wyner's result when different numbers of kernels are considered in the decoding process. Once this is done for the first phase, other phases can be similarly modified, to get a different approach to the feedback reliability function of the Poisson channel [6]. This approach might be more relevant in the neuroscience context. It would also be interesting to verify if the two approaches coincide.

References

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